

Year 7 • Chapter 11 • Probability: answers (teacher copy)

11.01 Sample space (booklet p. 3)

WE DO examples

- Ex 1: {H, T}
- Ex 2: {1, 2, 3, 4, 5, 6}
- Ex 3: {pass, fail}
- Ex 4: {G, PG, M, MA15+, R18+}
- Ex 5: 5; no, the sectors are different sizes
- Ex 6: {Katrina, Biljana}

Set A

- {A, B, C, D}
- {Mon, Tue, Wed, Thu, Fri, Sat, Sun}
- {a, e, i, o, u}
- {1, 2, 3, 4, 5, 6}
- {win, draw, lose}
- {summer, autumn, winter, spring}

Set B

- 5
- the sectors are different sizes
- 3
- 5
- 1

Set C

- e.g. {bus, train, ferry}: not equally likely
- {M, A, T, H, S}: equally likely
- {A, P, L, E}: P is more likely
- {0, 1, 2}: 1 is more likely
- {red, amber, green}: not equally likely
- {hearts, diamonds, clubs, spades}: equally likely

Set D

- {HH, HT, TH, TT}
- A-H, A-T, B-H, B-T, C-H, C-T, D-H, D-T
- 36 (6×6)
- e.g. sectors of 180° , 90° , 45° , 45°

Extension

- 16
- 9

Exit ticket

- Level 1: {red, yellow, blue}
- Level 2: 6
- Level 3: no: it depends on the student

11.02 Probability (booklet p. 7)

WE DO examples

- Ex 1: $\frac{1}{6}$
- Ex 2: $\frac{1}{2}$
- Ex 3: $\frac{2}{3}$
- Ex 4: $\frac{3}{10}$ (C)
- Ex 5: 20.2%
- Ex 6: 86.1%

Set A

- $\frac{1}{6}$
- $\frac{1}{2}$
- $\frac{2}{3}$
- 0
- 1
- $\frac{1}{3}$

Set B

- $\frac{2}{11}$
- $\frac{2}{11}$
- $\frac{4}{11}$
- $\frac{1}{11}$
- $\frac{4}{11}$
- 0

Set C

- $\frac{3}{10}$, 30.0%
- $\frac{3}{8}$, 37.5%
- $\frac{27}{40}$, 67.5%
- $\frac{21}{104}$, 20.2%
- $\frac{137}{208}$, 65.9%
- 0, 0%

Set D

- 0.44
- 1.6%
- 0.65
- 3

Extension

- $\frac{4}{13}$
- $\frac{3}{13}$

Exit ticket

- Level 1: $\frac{1}{6}$
- Level 2: $\frac{3}{10}$
- Level 3: 30%

11.03 The probability scale (booklet p. 11)

WE DO examples

- Ex 1: B
- Ex 2: unlikely
- Ex 3: rolling an odd or even number on a die (C)
- Ex 4: $\frac{3}{7}$
- Ex 5: $\frac{3}{14}$
- Ex 6: 0; 1

Set A

1. certain
2. impossible
3. even chance
4. very unlikely
5. likely
6. certain

Set B

1. impossible
2. unlikely
3. even chance
4. likely
5. certain
6. unlikely

Set C

1. $\frac{3}{7}$
2. $\frac{3}{14}$
3. 0
4. 1
5. $\frac{5}{14}$
6. $\frac{4}{7}$

Set D

1. A at $\frac{1}{2}$, B at $\frac{1}{6}$, C at 1, D at 0
2. own examples

Extension

1. 4
2. e.g. 2 blue, 7 red, 1 green

Exit ticket

1. Level 1: 0
2. Level 2: very likely
3. Level 3: $\frac{5}{8}$

11.04 Experimental probability (booklet p. 15)**WE DO examples**

1. Ex 1: $\frac{1}{2}$
2. Ex 2: 40
3. Ex 3: $\frac{3}{10}$
4. Ex 4: fewer heads than expected (24 vs 40)
5. Ex 5: 10
6. Ex 6: the results even out over many trials

Set A

1. $\frac{1}{2}$
2. $\frac{1}{6}$
3. $\frac{1}{4}$
4. $\frac{1}{2}$
5. $\frac{3}{5}$
6. $\frac{2}{5}$

Set B

1. 40
2. 10
3. 25
4. 50
5. 60
6. 500

Set C

1. $\frac{3}{10}$
2. $\frac{1}{4}$
3. $\frac{21}{50}$
4. $\frac{7}{20}$
5. $\frac{13}{40}$
6. 0

Set D

1. own results
2. answers vary; class results close to $\frac{1}{2}$

Extension

1. about 150
2. no: red has $P \approx 0.3$, not 0.25

Exit ticket

1. Level 1: $\frac{1}{6}$
2. Level 2: 20
3. Level 3: $\frac{5}{24}$

11.05 Complementary events (booklet p. 19)**WE DO examples**

1. Ex 1: not a queen, $\frac{12}{13}$
2. Ex 2: not a multiple of 3, $\frac{2}{3}$
3. Ex 3: not winning, $\frac{549}{550}$
4. Ex 4: $\frac{1}{5}$
5. Ex 5: $\frac{3}{5}$; $\frac{3}{5}$
6. Ex 6: $\frac{7}{15}$

Set A

1. not rolling a 6
2. getting a tail
3. drawing a black card
4. losing
5. choosing a consonant
6. rolling an odd number

Set B

1. $\frac{3}{4}$
2. 0.7
3. 55%
4. $\frac{7}{12}$
5. 0.93
6. 0

Set C

1. $\frac{12}{13}$
2. $\frac{2}{3}$
3. $\frac{549}{550}$
4. $\frac{4}{5}$
5. $\frac{7}{15}$
6. $\frac{3}{4}$

Set D

1. 0.65
2. 0.3
3. 8
4. an event either happens or it does not

Extension

1. $\frac{5}{12}$
2. 130

Exit ticket

1. Level 1: not rolling a 2
2. Level 2: 0.2
3. Level 3: $\frac{1}{2}$